Light

GCSE Computer Science - Fetch-Decode-Execute Cycle and Embedded Systems (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which stage of the fetch-decode-execute cycle must always happen first?

- ○ **A.** Decode
- ○ **B.** Execute
- ○ **C.** Store
- ○ **D.** Fetch

Q2 (1 mark)

What is the output of the decode stage?

- ○ **A.** The numerical result of a calculation

- ○ **B.** A copy of the instruction in secondary storage
- ○ **C.** An understanding of what operation to perform and what data is needed
- ○ **D.** A printed document

Q3 (1 mark)

During the execute stage, which operations might involve accessing main memory?

- ○ **A.** Memory is never accessed during execute
- ○ **B.** Only calculations require memory access
- ○ **C.** Reading data from a memory location or writing results to a memory location
- ○ **D.** Only the monitor requires memory access

Q4 (1 mark)

Which of these is an example of a logical operation performed by the ALU?

- ○ **A.** Adding two numbers together
- ○ **B.** Dividing one number by another
- ○ **C.** Checking whether one value is greater than another
- ○ **D.** Storing data in memory

Q5 (1 mark)

What role does the control unit play during instruction decoding?

- ○ **A.** It interprets the binary instruction to determine what action is required
- ○ **B.** It performs mathematical calculations
- ○ **C.** It stores the decoded instruction permanently
- ○ **D.** It connects the computer to the network

Q6 (1 mark)

What does each pulse from the CPU clock enable?

- ○ **A.** The computer to connect to the internet
- ○ **B.** The screen brightness to increase
- ○ **C.** The hard drive to spin faster
- ○ **D.** One or more steps of processing to occur in a synchronised manner

Q7 (1 mark)

What is special about the data bus compared to other buses?

- ○ **A.** It can carry data in both directions between components
- ○ **B.** It only operates when the computer is switched off

- ○ **C.** It is only used for audio output
- ○ **D.** It stores programs permanently

Q8 (1 mark)

What type of data does cache memory typically store?

- ○ **A.** Only video files
- ○ **B.** Internet browsing history only
- ○ **C.** Permanent system settings
- ○ **D.** Frequently and recently accessed data and instructions from main memory

Q9 (1 mark)

How does the access speed of registers compare to other memory types?

- ○ **A.** Registers are the slowest memory in the computer
- ○ **B.** Registers are the fastest memory, faster than cache and RAM
- ○ **C.** Registers are slower than RAM but faster than hard drives
- ○ **D.** All memory types have identical access speeds

Q10 (1 mark)

What is typically true about the software in an embedded system?

- ○ **A.** Users can freely install any applications they choose
- ○ **B.** The software is fixed by the manufacturer for a specific purpose
- ○ **C.** Embedded systems do not contain any software
- ○ **D.** The software must be reinstalled daily

Q11 (1 mark)

Which everyday device contains an embedded system?

- ○ **A.** A microwave oven
- ○ **B.** A wooden desk
- ○ **C.** A standard light bulb without smart features
- ○ **D.** A paper notebook

Q12 (1 mark)

What does it mean for processor cores to work in parallel?

- ○ **A.** The cores take turns working one at a time
- ○ **B.** Parallel processing means cores are physically arranged in lines
- ○ **C.** The cores must all run the same instruction together
- ○ **D.** Multiple cores process different instructions simultaneously

Q13 (2 marks)

Why is the fetch-decode-execute cycle described as a cycle rather than a sequence?

- ○ **A.** Because it only happens once when the computer starts
- ○ **B.** Because the stages can occur in any order
- ○ **C.** Because after execute completes, the process returns to fetch for the next instruction, repeating continuously
- ○ **D.** Because the same instruction is processed repeatedly forever

Q14 (2 marks)

A student confuses the roles of the ALU and control unit. How would you explain the key difference between them?

- ○ **A.** The control unit coordinates and directs operations while the ALU performs the actual calculations and comparisons
- ○ **B.** There is no difference; they are two names for the same component
- ○ **C.** The ALU controls the clock speed while the control unit stores data
- ○ **D.** Both components only work when the computer is switched off

Q15 (2 marks)

If a CPU's clock speed doubled from 2 GHz to 4 GHz with all else equal, what would be the theoretical effect?

- ○ **A.** The CPU would use half as much power

- ○ **B.** There would be no change in performance
- ○ **C.** The CPU's storage capacity would double
- ○ **D.** The CPU could potentially process twice as many instructions per second

Q16 (2 marks)

Why do embedded systems typically start operating almost instantly when powered on?

- ○ **A.** They contain no software at all
- ○ **B.** They have faster internet connections
- ○ **C.** They run simple, dedicated software stored in ROM or flash memory that does not require a complex boot process like general-purpose operating systems
- ○ **D.** Embedded systems never need to be powered on

Q17 (2 marks)

A computer has a 32-bit address bus. What does this determine?

- ○ **A.** How fast data can be transferred
- ○ **B.** The maximum amount of memory that can be directly addressed by the CPU
- ○ **C.** How many programs can run simultaneously
- ○ **D.** The physical length of the wires

Q18 (2 marks)

Why would the CPU use a register to hold the result of a calculation rather than writing it directly to RAM?

- ○ **A.** Using a register is much faster, and the result may be needed immediately for the next calculation
- ○ **B.** RAM cannot store numbers
- ○ **C.** Registers have unlimited storage capacity
- ○ **D.** RAM is only used for video games

Q19 (2 marks)

Why can embedded systems often be more thoroughly tested than general-purpose software?

- ○ **A.** Embedded systems are never tested
- ○ **B.** General-purpose software has fewer features to test
- ○ **C.** Testing is illegal for embedded systems
- ○ **D.** They have limited, well-defined functions, making it possible to test all expected behaviours systematically

Q20 (2 marks)

What happens when a cache miss occurs?

- ○ **A.** The computer immediately shuts down
- ○ **B.** The CPU permanently loses the ability to access that data

- ○ **C.** The required data must be fetched from slower main memory, causing a delay
- ○ **D.** Cache misses improve computer performance

Q21 (2 marks)

Some users 'overclock' their CPUs by increasing clock speed beyond manufacturer specifications. What risk does this create?

- ○ **A.** The CPU will immediately become more reliable
- ○ **B.** Excessive heat generation may damage the CPU or cause system instability
- ○ **C.** Overclocking makes the computer use less electricity
- ○ **D.** The CPU will store more data

Q22 (2 marks)

How does an operating system utilise multiple processor cores?

- ○ **A.** It assigns different processes or threads to different cores, allowing genuine multitasking
- ○ **B.** Operating systems cannot use multiple cores
- ○ **C.** All cores must run the same program simultaneously
- ○ **D.** Only one core is ever active at any time

Q23 (2 marks)

What challenge exists when embedded system software needs to be updated?

- ○ **A.** Embedded systems can be updated as easily as smartphone apps
- ○ **B.** Software updates are not possible for any electronic device
- ○ **C.** Embedded systems automatically update themselves every hour
- ○ **D.** Updates may require specialised equipment, physical access, or careful testing since failures could render the device non-functional

Q24 (3 marks)

Modern CPUs use a technique called pipelining where they begin fetching the next instruction before the current one finishes executing. Analyse how this improves efficiency compared to waiting for each instruction to complete before starting the next.

- ○ **A.** Pipelining has no effect on CPU efficiency
- ○ **B.** Pipelining reduces efficiency because the CPU becomes confused by multiple instructions
- ○ **C.** Pipelining increases throughput by overlapping stages - while one instruction executes, another decodes, and another fetches, keeping all parts of the CPU busy simultaneously
- ○ **D.** Pipelining only works for computers without any memory

Q25 (3 marks)

Trace the complete process when a program needs to store the value 42 in memory location 5000, identifying all CPU components and buses involved.

○ **A.** The control unit decodes the store instruction and coordinates the operation; the value 42 travels from a register to the data bus; address 5000 is placed on the address bus; the control bus signals a write operation; memory stores the value at the specified location

○ **B.** Only the ALU is involved in storing data

○ **C.** Memory stores data without any CPU involvement

○ **D.** Data is stored by physically moving the CPU closer to the memory location

Q26 (3 marks)

A computer has a very fast CPU (5 GHz, 16 cores) but slow RAM and a mechanical hard drive. Analyse why this system might still perform poorly despite its powerful processor.

○ **A.** CPU specifications are the only factor affecting performance

○ **B.** The CPU will be frequently waiting for data from slow memory and storage; no matter how fast the processor, it cannot process data faster than memory can supply it, creating a bottleneck that wastes processing capacity

○ **C.** Faster CPUs always make all other components work faster automatically

○ **D.** Memory and storage speed have no effect on overall system performance

Q27 (3 marks)

A modern car contains over 100 embedded systems. Analyse why automotive manufacturers use many separate embedded systems rather than one powerful central computer.

- ○ **A.** Using many systems is simply a manufacturing error
- ○ **B.** Cars do not actually contain any computer systems
- ○ **C.** One central computer would be cheaper and more reliable
- ○ **D.** Individual systems provide fault isolation (one failure does not crash everything), allow specialised optimisation for each function, enable different teams to develop subsystems independently, and simplify certification since each system's limited function can be verified thoroughly

Q28 (3 marks)

Explain why computer systems implement multiple levels of cache (L1, L2, L3) rather than a single cache layer.

- ○ **A.** Multiple levels exist only for marketing purposes with no technical benefit
- ○ **B.** L1 cache is extremely fast but tiny and expensive; L2 is slower but larger; L3 is slower still but even larger. This hierarchy provides fast access for the most critical data while extending cache benefits to more data than a single level could practically provide
- ○ **C.** All cache levels have identical speed and capacity
- ○ **D.** Only one cache level is ever used at a time

Q29 (3 marks)

During normal program execution, an interrupt signal arrives (for example, a key is pressed). Analyse how this affects the fetch-decode-execute cycle.

- ○ **A.** The CPU completes the current instruction, saves its current state, jumps to an interrupt handler routine to process the event, then

restores state and resumes the original program's FDE cycle

○ **B.** Interrupts are ignored and the current program continues unaffected

○ **C.** Interrupts permanently stop all processing

○ **D.** The keyboard directly changes values in the program's memory

Q30 (3 marks)

Internet of Things (IoT) devices like smart sensors and connected appliances are embedded systems with network connectivity. Analyse the unique challenges this combination creates.

○ **A.** IoT devices have no unique challenges compared to traditional embedded systems

○ **B.** IoT devices must balance embedded system constraints (low power, limited resources, dedicated function) with networking requirements (security updates, communication protocols, protection against cyber attacks), creating complex design challenges

○ **C.** Network connectivity makes IoT devices identical to desktop computers

○ **D.** IoT devices do not need any security because they are embedded systems

Total: **55** marks • Time: **30 min**

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